

STUDENT NUMBER: _____

TEACHER NAME: _____

BAULKHAM HILLS HIGH SCHOOL

MATHEMATICS ADVANCED ASSESSMENT

December 2009

Time allowed – Fifty minutes
Plus 5 minutes reading time

DIRECTIONS TO CANDIDATES:

- Show all relevant working.
- Use black or blue pen.
- NO liquid paper is to be used.
- Approved Maths aids and calculators may be used.

QUESTION 1 [6 marks]

Find the primitive function of:

(i.) $2x^2 + 5x - 1$

2

(ii.) $\frac{x^2 + 2}{x^2}$

2

(iii.) $(2x + 5)^{10}$

2

QUESTION 2 [3 marks]Redraw and complete the table below for the function $y = x\sqrt{4-x}$ correct to 4 decimal places.

x	0	1	2	3	4
y	0			3	0

1

Hence estimate $\int_0^4 x\sqrt{4-x} dx$ using Simpson's rule with 5 function values correct to 3 decimal places.

2

QUESTION 3 [4 marks]Find the vertex, focal length, focus and directrix of $x^2 - 6x + y + 18 = 0$.

4

QUESTION 4 [5 marks](a) Find the area bounded by $y = x^2 - 6x + 8$, the x axis and the lines $x = 0$ and $x = 4$.

3

(b) If $\int_0^k 4 - 2x dx = 4$, find k .

2

QUESTION 5 [4 marks]If $y = 2x\sqrt{x+1}$

(i) Show that $\frac{dy}{dx} = \frac{3x+2}{\sqrt{x+1}}$

2

(ii) Hence evaluate $\int_3^8 \frac{3x+2}{\sqrt{x+1}} dx$.

2

QUESTION 6 [4 marks]

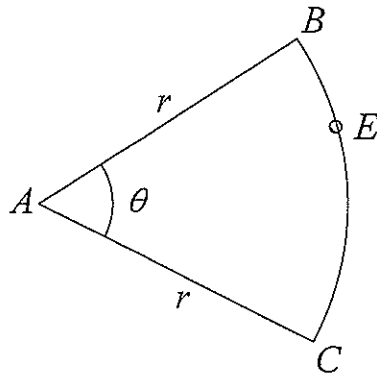
Sketch $y = 1 + \cos 2x$ for $0 \leq x \leq 2\pi$.
State the period and amplitude.

4

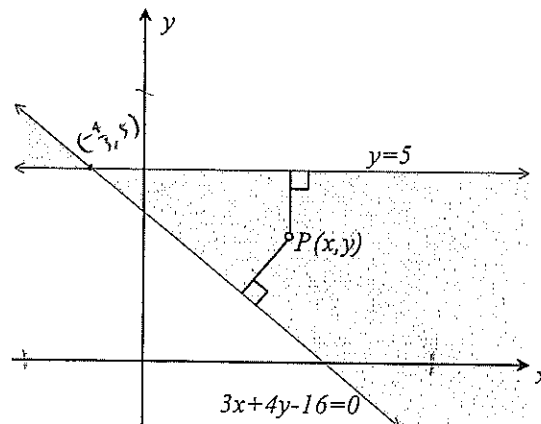
QUESTION 7 [4 marks]

In the figure AB and AC are radii of a circle with centre A . $AB = AC = r$ metres.
 E lies on arc BC .

4



If the perimeter of the figure $ABEC$ is 16 metres, find the area Y (in m^2) of the sector $ABEC$. Answer in simplest form in terms of r .

QUESTION 8 [4 marks]

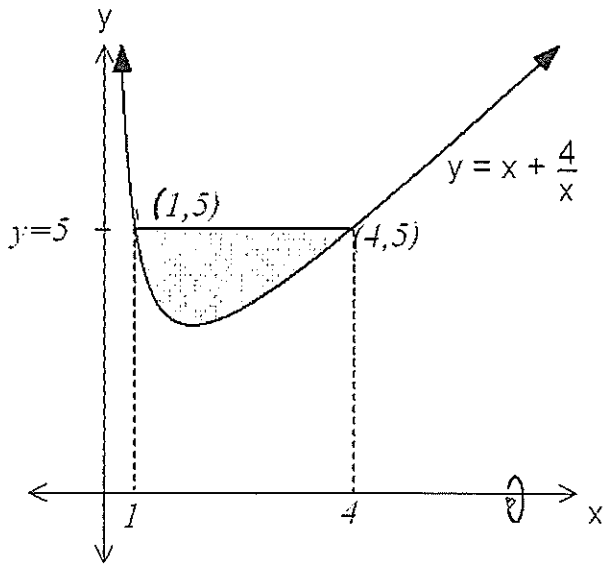
The point $P(x, y)$ is equidistant from the lines $y = 5$ and $3x + 4y - 16 = 0$ and lies in the shaded region of the diagram. Find the equation of the locus of P .

4

QUESTION 9 [4 marks] (Start on a new page)

Given $y = x + \frac{4}{x}$.

4



The shaded region is rotated around the x – axis. Find the volume of the solid of revolution in terms of π .

END OF EXAMINATION

$$\textcircled{1} \textcircled{a) \int 2x^2 + 5x - 1 dx} \quad \textcircled{1} \textcircled{P}$$

$$= 2 \frac{x^3}{3} + 5 \frac{x^2}{2} - x + C$$

$$\textcircled{ii) \int \frac{x^2+2}{x^2} dx} = \int 1 + 2x^{-1} dx \quad \textcircled{1}$$

$$= x + \frac{2x^{-1}}{-1} + C = x - \frac{2}{x} + C \quad \text{OR}$$

$$\textcircled{iii) \int (2x+5)^{10} dx} = \frac{(2x+5)^{11}}{11 \times 2} + C \quad \textcircled{1}$$

x	0	1	2	3	4
y	0	$\sqrt{3}$	$2\sqrt{2}$	3	0
		1.7321	2.8284	3	0

$$y = x\sqrt{4-x} \quad \textcircled{1}$$

$$A = \frac{1}{3} (0 + 0 + 4(\sqrt{3} + 3) + 2 \times 2\sqrt{2}) \quad \textcircled{1}$$

$$= \frac{1}{3} \times 24.5852 = \underline{8.195} \quad \textcircled{1}$$

$$\textcircled{3) x^2 - 6x + y + 18 = 0}$$

$$x^2 - 6x = -y - 18$$

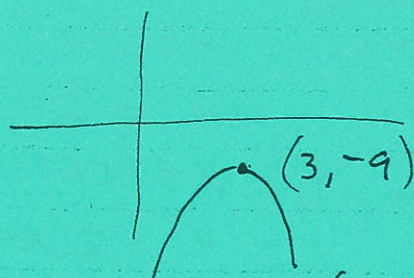
$$x^2 - 6x + 3^2 = -y - 18 + 9$$

$$(x-3)^2 = -(y+9) \quad \textcircled{1}$$

vertex $(3, -9)$ $\textcircled{1}$

focal length = $a = \frac{1}{4}$ $\textcircled{1}$

$$4a = 1$$



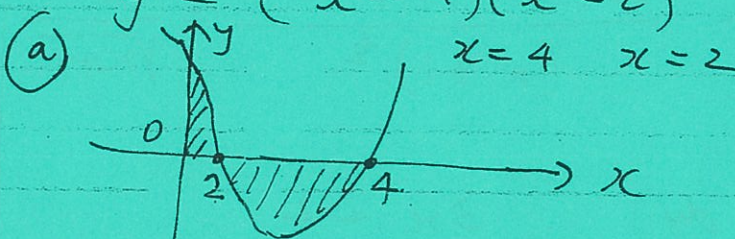
Focus $S(3, -9\frac{1}{4})$ $\textcircled{1}$

directrix $y = -8\frac{3}{4}$ $\textcircled{1}$

$$\textcircled{4) y = x^2 - 6x + 8}$$

$$y = (x-4)(x-2)$$

$x=4 \quad x=2$



$$A = \int_0^2 x^2 - 6x + 8 dx + \int_2^4 x^2 - 6x + 8 dx \quad \textcircled{1}$$

$$= \left[\frac{x^3}{3} - \frac{6x^2}{2} + 8x \right]_0^2 + \left[\frac{x^3}{3} - 3x^2 + 8x \right]_2^4 \quad \textcircled{1}$$

$$= \frac{8}{3} - 3 \times 4 + 16 - 0 + \left[\frac{4^3}{3} - 3 \times 16 + 32 \right] - \left(\frac{8}{3} - 3 \times 4 + 16 \right) \quad \textcircled{1}$$

$$= \frac{20}{3} + \left| -\frac{4}{3} \right| = \underline{8} \quad \textcircled{1}$$

$$\textcircled{b) \int_0^k 4 - 2x dx} = 4$$

$$\left[4x - x^2 \right]_0^k = 4 \quad \textcircled{1}$$

$$4k - k^2 = 4 \quad \therefore \textcircled{k=2} \quad \textcircled{1}$$

$$0 = k^2 - 4k + 4$$

⑤ $y = 2x\sqrt{x+1}$ $u = 2x$
 $u' = 2$
 $v = (x+1)^{\frac{1}{2}}$
 $v' = \frac{1}{2}(x+1)^{-\frac{1}{2}}$

$$\frac{dy}{dx} = u'v + v'u$$

$$= 2(x+1)^{\frac{1}{2}} + \frac{1}{2}(x+1)^{-\frac{1}{2}} \cdot 2x$$

$$= \frac{2(x+1)^{\frac{1}{2}}}{1} + \frac{x}{(x+1)^{\frac{1}{2}}}$$

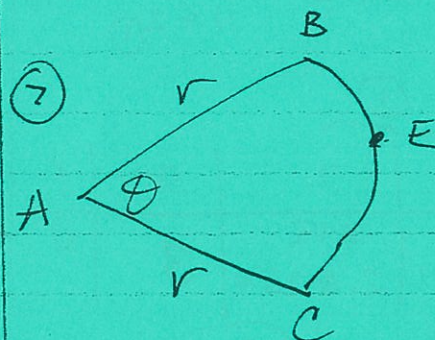
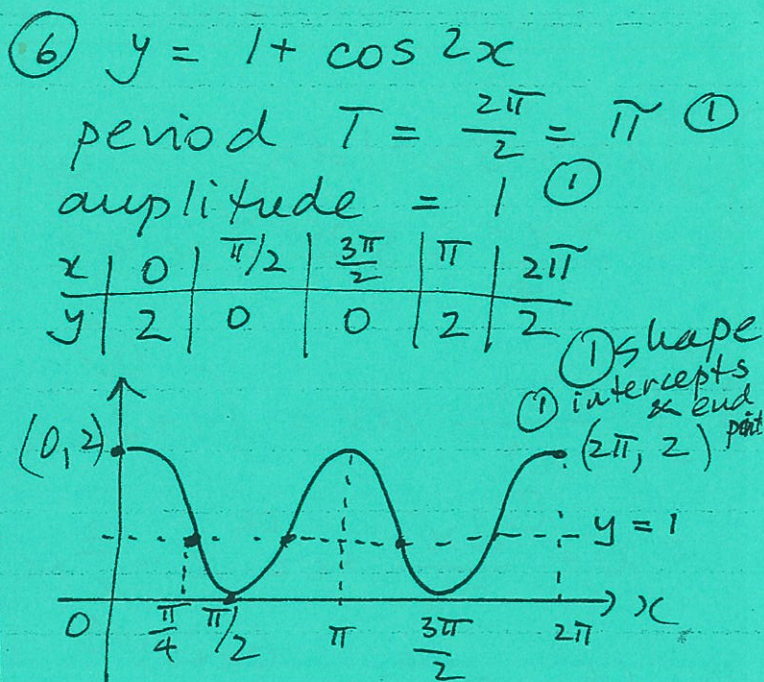
$$= \frac{2(x+1) + x}{(x+1)^{\frac{1}{2}}} = \frac{3x+2}{\sqrt{x+1}}$$

ii) $\frac{dy}{dx} = \frac{3x+2}{\sqrt{x+1}}$

$$y = \int \frac{3x+2}{\sqrt{x+1}} dx$$

$$\therefore \int_3^8 \frac{3x+2}{\sqrt{x+1}} dx = \left[2x\sqrt{x+1} \right]_3^8$$

$$= [16\sqrt{9} - 6\sqrt{4}] = 36$$



$$P = 16$$

$$P = 2r + r\theta = 16$$

$$r\theta = 16 - 2r$$

$$\theta = \frac{16-2r}{r}$$

$$\text{Area} = \frac{1}{2}r^2\theta = \frac{1}{2}r^2 \times \frac{16-2r}{r}$$

$$\therefore A = \frac{1}{2}r \frac{2(8-r)}{1} = r(8-r)$$

$$\text{or } \boxed{A = 8r - r^2} \quad \text{①}$$

8) $P(x, y)$

$y = 5$
 $y - 5 = 0$

$3x + 4y - 16 = 0$

$$\frac{|y - 5|}{\sqrt{0^2 + 1^2}} = \frac{|3x + 4y - 16|}{\sqrt{3^2 + 4^2}}$$

$$|y - 5| = |3x + 4y - 16|$$

i) $5(y - 5) = 3x + 4y - 16$
 $0 = 3x + 4y - \sqrt{5}y - 16 + 5\sqrt{5}$
 $0 = 3x - y + 9 \therefore y = 3x + 9$
 OR

ii) $-5(y - 5) = 3x + 4y - 16$
 $-5y + 25 = 3x + 4y - 16$
 $-9y = 3x - 41$
 $y = -\frac{3}{9}x - \frac{41}{9}$

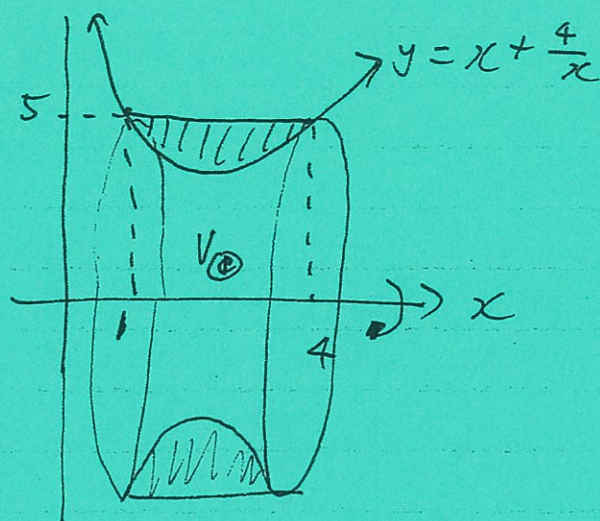
locus of P is

i) $y = 3x + 9$
 or ii) $y = -\frac{1}{3}x - \frac{41}{9}$

if it has to lie in shaded region
 $\therefore$ gradient must be negative

$\therefore y = -\frac{1}{3}x - \frac{41}{9}$
 or $0 = 3x + 9y - 41$

9) $y = x + \frac{4}{x}$



$$V = V_{\text{cylinder}} - V_0$$

$$= \pi r^2 \times h - \pi \int_1^4 \left(x + \frac{4}{x}\right)^2 dx$$

V_0 = volume under $y = x + \frac{4}{x}$

$$\therefore V_0 = \pi \int_1^4 \left(x + \frac{4}{x}\right)^2 dx$$

$$= \pi \int_1^4 \left(x^2 + 8 + \frac{16}{x^2}\right) dx$$

$$= \pi \left[\frac{x^3}{3} + 8x + \frac{16x^{-1}}{-1} \right]_1^4$$

$$= \pi \left[\frac{4^3}{3} + 32 - \frac{16}{4} - \left(\frac{1}{3} + 8 - 16\right) \right]$$

$$V_0 = 57\pi$$

$r = 5$ $h = 3$

$$V_{\text{cylinder}} = \pi \times 5^2 \times 3 = 75\pi$$

$$\therefore V = 75\pi - 57\pi = 18\pi$$